INCH-POUND
MIL-DTL-24643/31E
w/AMENDMENT 1
03 June 2005
SUPERSEDING
MIL-DTL-24643/31E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SU (INCLUDING VARIATION LS2SUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in three variations: LS2SU and LS2SUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First	-	Copper conductor, uncoated, ASTM B286, AWG size No. 22, 7 strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Cross-linked polyethylene insulation. Nominal wall thickness 0.013 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
Fourth	-	Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches to form a pair.
Fifth	-	Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
Sixth	-	Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material, 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.
Eighth	-	The required number of pairs (see Table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.
Ninth	-	Binder tape, applied helically with overlap.

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Construction variant for type LS2SU

Tenth - Cross-linked polyolefin jacket. (See Table I for thickness.)

Construction variant for type LS2SUS

Tenth - Braided shield of tin coated copper.

Eleventh - Spirally wrapped 0.002-inch thick polyethylene (polyester film) separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of tin coated copper.

Thirteenth - Binder tape applied helically with overlap.

Fourteenth - Cross-linked polyolefin jacket. (See Table I for thickness.)

TABLE I. Details.

Military part no.1/ M24643/31	Size2/	Number of pairs	Cable jacket thickness min avg (inch)	Diameter over shielded core (inch nom)	Overall diameter (inches)			
					TYPE LS2SU		TYPE LS2US	
					min	max	min	max
-01UN	3	3	0.050	0.399	0.480	0.520	0.540	0.580
-02UN	7	7	0.050	0.531	0.610	0.660	0.670	0.720
-03UN	10	10	0.050	0.617	0.770	0.830	0.830	0.890
-04UN	14	14	0.063	0.756	0.860	0.930	0.920	0.990
-05UN	19	19	0.063	0.796	0.970	1.040	1.030	1.100
-06UN	24	24	0.063	0.885	1.120	1.210	1.180	1.270
-07UN	30	30	0.063	0.982	1.190	1.280	1.250	1.340
-08UN	37	37	0.063	1.030	1.290	1.380	1.350	1.440
-09UN	44	44	0.063	1.198	1.460	1.550	1.520	1.610
-10UN	61	61	0.075	1.262	1.660	1.740	1.720	1.800

1/ When double overall shield is required, see MIL-DTL-24643 for configuration of part number.

2/ Type LS2SU and LS2SUS.

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25 °C, maximum.....17.37

Voltage withstand - volts, root mean square, minimum

Conductor to conductor.....2000

Conductor to shield.....1000

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Shield to shield (pair shields only)..... 500

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Insulation resistance - megohms - 1000 feet minimum

Conductor to conductor..... 500

Component shield to shield..... 100

Shield to shield (LS2SUS only)..... 100

Jacket flaws (LS2SUS only)..... No failure

Conductor and shield continuity..... No failure

Group A:

Visual and dimensional..... No failure

Capacitance

Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum..... 30

Capacitance unbalance - percent maximum..... 8

Characteristic impedance - at 1 MHz, ohms..... 75 + 5

Group B:

Cross-linked proof test (percent, maximum)

Insulation..... 50

Jacket (When tested at 200 °C)..... 50

Attenuation - at 3 MHz, decibels per 100 FT, maximum..... 3

Tear - pounds per inch thickness, minimum (ASTM D470)..... 35

Physicals (unaged)

Insulation

Tensile strength - lb/in^2, minimum..... 1800

Elongation - percent, minimum..... 250

Jacket (cable)

Tensile strength - lb/in^2, minimum..... 1300

Elongation - percent, minimum..... 160

Group C:

Physicals (aged)

Insulation

Air oven

Tensile strength - percent of unaged, minimum..... 80

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Elongation - percent of unaged minimum 80

Jacket (cable)

Air oven

Tensile strength - percent of unaged, minimum 60

Elongation - percent of unaged minimum 60

Hot oil immersion

Tensile strength - percent of unaged, minimum 50

Elongation - percent of unaged minimum 50

Shrinkage ..... No failure

Heat distortion - percent of unaged, maximum 30

Permanence of printing (jacket) - cycles, minimu m 125

Shield - conformance to material, construction and cover ..... No failure

Surface transfer impedance (type LSMUS only)

Milliohms per meter, maximum 700

EMP response time - dB, minimum 60

Group D:

Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:

Aging and compatibility (cable) (125 + 5 °C) ..... No failure

Abrasion resistance (jacket) - scrapes, minimum 75

Acid gas equivalent - percent, maximum

Jacket 2

Fillers 2

Insulation 18

Shield insulation 2

Halogen content - percent, maximum

Jacket 0.2

Fillers 0.2

Immersion (jacket)

Tensile strength - percent of unaged, minimum 50

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Elongation - percent of unaged, minimu m 50

Smoke index, maximum

Jacket..... 25

Fillers 45

Insulation..... 45

Toxicity index, maximum

Jacket 5

Fillers 5

Insulation..... 1.5

Durometer (jacket) - (type A) hardness, minimum..... 80

Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 30	1000
37 through 61	500

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2005-011)

Review Activities:
Army – AV, CR
Navy – CG, EC
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